

Nature Machine Intelligence

A Feynman Path Integral Framework for Adaptive Anastomotic Patterning via Finite Mathematics

Authors: Cartik Sharma et al.

Abstract

This study introduces a novel non-rigid registration paradigm governing the surgicobotanical application of 'surfboard' excisions in complex anastomosis. We extend traditional Coherent Point Drift by formulating the deformation field as a quantum-like system governed by Feynman Path Integrals. The action functional is mapped to finite properties of numbers and prime algebraic structures, effectively regularizing structural alignment over multiple simulated geometrical histories to achieve an optimal manifold mapping.

1. Feynman Path Integrals (Sum Over Histories)

The non-rigid deformation from the source lattice X to the target Y is evaluated across all possible stochastic trajectories $x(t)$. The transition amplitude K is the integral over all paths, weighted by the action $S[x(t)]$:

$$K(X_f, t_f; X_i, t_i) = \int \mathcal{D}[x(t)] e^{-S[x(t)]/\gamma}$$

Here, γ is a regularization parameter analogous to Planck's constant, and $\mathcal{D}[x(t)]$ represents the measure over the space of deformations. The action functional $S[x(t)]$ encodes the geometric residual error:

$$S[x(t)] = \int_{t_i}^{t_f} \left(\frac{1}{2} \alpha \|\dot{x}(t)\|^2 + V(x(t), Y) \right) dt$$

2. Finite Math Coupling for Residual Optimization

To constrain the functional space to biological equivalences, the action S is partitioned into finite fields using equivalence properties of prime residuals. The residual equation governing the potential V becomes:

$$V(x_k, Y) \equiv \min_{y \in Y} \|x_k - y\|_W^2 \pmod{p_m}$$

where p_m iterates over a set of anatomical characteristic primes $\mathcal{P}$. The differential updating rule combines the Feynman weight $w_p = e^{-S_p}$ with a numerical phase factor bridging discrete geometries:

$$x^{(i+1)} = \frac{\sum_{paths} x_{path}^{(i)} \cdot e^{-S_{path}} \cos(2\pi/p_m)}{\sum_{paths} e^{-S_{path}}}$$

Conclusion

By mapping the continuous Feynman Path Integral to finite number fields, we establish a highly robust and globally optimal alignment algorithm for surgical anastomosis.

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